

## GUANGZHOU, China

WMO: 592870

Lat: 23.2099N

Lon: 113.4822E

Elev: 72

StdP: 100.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.9        | 7.0 | -7.1                          | 2.1 | 11.7 | -4.0 | 2.7 | 12.1 | 8.3                   | 11.6 | 7.4 | 11.7 | 3.3                   | 0    | 0.332 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.6                    | 36.0            | 26.2 | 35.0 | 26.2 | 34.0 | 26.1 | 27.8                | 31.8 | 27.5 | 31.4 | 27.1 | 31.0 | 2.8                  | 340  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.0                            | 22.9 | 29.8 | 26.3 | 21.9 | 29.3 | 26.1 | 21.7 | 29.2 | 89.8          | 32.4 | 88.2 | 31.8 | 86.8 | 31.4 | 29.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 7.4               | 6.3  | 5.4 | DB                         | 3.7 | 37.7               | 1.6 | 1.1                                                | 2.5  | 38.5       | 1.6  | 39.1       | 0.8  | 39.8       | -0.4 | 40.6 |  |
|                   |      |     | WB                         | 0.9 | 28.7               | 1.7 | 0.4                                                | -0.4 | 29.0       | -1.4 | 29.2       | -2.4 | 29.5       | -3.6 | 29.8 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 23.2   | 14.5  | 16.1  | 18.8  | 23.2  | 26.6  | 28.7  | 29.7  | 29.5  | 28.2  | 25.3  | 20.9  | 16.1 |
|                                                                               | DBStd       | 6.21   | 3.93  | 4.71  | 3.90  | 3.30  | 2.35  | 1.83  | 1.69  | 1.63  | 2.03  | 2.38  | 3.31  | 3.40 |
|                                                                               | HDD10.0     | 16     | 8     | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    |
|                                                                               | HDD18.3     | 371    | 129   | 91    | 44    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 14    | 87   |
|                                                                               | CDD10.0     | 4818   | 147   | 176   | 273   | 395   | 514   | 560   | 609   | 604   | 546   | 476   | 328   | 191  |
|                                                                               | CDD18.3     | 2130   | 11    | 27    | 58    | 151   | 256   | 310   | 351   | 346   | 296   | 217   | 92    | 17   |
|                                                                               | CDH23.3     | 22007  | 29    | 100   | 235   | 996   | 2443  | 3589  | 4421  | 4312  | 3391  | 1903  | 524   | 63   |
|                                                                               | CDH26.7     | 8872   | 1     | 7     | 22    | 228   | 839   | 1493  | 2096  | 2020  | 1444  | 619   | 101   | 3    |
| Wind                                                                          | WSAvg       | 2.4    | 2.5   | 2.4   | 2.3   | 2.2   | 2.2   | 2.3   | 2.4   | 2.1   | 2.3   | 2.4   | 2.5   | 2.7  |
| Precipitation                                                                 | PrecAvg     | 1880   | 56    | 57    | 95    | 193   | 296   | 348   | 237   | 257   | 190   | 69    | 44    | 37   |
|                                                                               | PrecMax     | 2820   | 430   | 222   | 262   | 400   | 765   | 881   | 402   | 476   | 525   | 256   | 168   | 145  |
|                                                                               | PrecMin     | 1277   | 1     | 0     | 22    | 28    | 129   | 144   | 50    | 42    | 62    | 1     | 1     | 0    |
|                                                                               | PrecStd     | 381    | 79    | 53    | 62    | 102   | 148   | 153   | 92    | 115   | 105   | 71    | 44    | 39   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.9  | 27.9  | 28.8  | 32.0  | 34.9  | 36.0  | 37.2  | 37.1  | 35.9  | 34.0  | 30.9  | 27.1 |
|                                                                               |             | MCWB   | 18.9  | 21.0  | 22.0  | 23.8  | 25.8  | 26.5  | 26.4  | 26.3  | 25.9  | 23.9  | 22.2  | 20.0 |
|                                                                               | 2%          | DB     | 24.0  | 26.0  | 27.1  | 30.5  | 33.0  | 34.5  | 36.0  | 35.9  | 34.8  | 32.1  | 29.1  | 25.1 |
|                                                                               |             | MCWB   | 17.9  | 19.7  | 21.1  | 23.7  | 25.4  | 26.4  | 26.4  | 26.1  | 25.7  | 23.0  | 21.2  | 18.4 |
|                                                                               | 5%          | DB     | 22.2  | 24.4  | 25.9  | 29.1  | 32.0  | 33.5  | 34.9  | 34.9  | 33.7  | 31.0  | 27.9  | 23.5 |
|                                                                               |             | MCWB   | 16.8  | 19.0  | 20.7  | 23.3  | 25.0  | 26.3  | 26.5  | 26.2  | 25.4  | 22.6  | 20.5  | 17.5 |
|                                                                               | 10%         | DB     | 20.9  | 23.0  | 24.5  | 28.0  | 30.9  | 32.3  | 33.8  | 33.8  | 32.4  | 29.9  | 26.2  | 21.9 |
|                                                                               |             | MCWB   | 16.3  | 18.7  | 20.3  | 23.0  | 24.8  | 26.2  | 26.5  | 26.2  | 24.9  | 22.2  | 19.6  | 15.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.6  | 22.7  | 23.8  | 26.1  | 27.7  | 28.2  | 28.3  | 28.2  | 27.7  | 26.1  | 24.3  | 22.5 |
|                                                                               |             | MCDB   | 23.4  | 25.4  | 26.4  | 29.6  | 31.2  | 32.1  | 32.8  | 32.4  | 32.0  | 30.2  | 28.0  | 24.8 |
|                                                                               | 2%          | WB     | 19.4  | 21.3  | 23.0  | 25.2  | 26.9  | 27.7  | 27.7  | 27.7  | 27.0  | 25.4  | 23.3  | 20.6 |
|                                                                               |             | MCDB   | 22.5  | 24.1  | 25.3  | 28.3  | 30.4  | 31.3  | 31.9  | 31.7  | 31.2  | 29.2  | 26.8  | 23.5 |
|                                                                               | 5%          | WB     | 18.3  | 20.3  | 22.1  | 24.6  | 26.4  | 27.2  | 27.4  | 27.2  | 26.6  | 24.7  | 22.5  | 18.8 |
|                                                                               |             | MCDB   | 21.5  | 23.1  | 24.6  | 27.4  | 29.7  | 30.7  | 31.5  | 31.3  | 30.6  | 28.4  | 25.9  | 22.0 |
|                                                                               | 10%         | WB     | 16.9  | 19.2  | 21.2  | 24.0  | 26.0  | 26.8  | 27.0  | 26.8  | 26.2  | 24.1  | 21.5  | 17.2 |
|                                                                               |             | MCDB   | 20.0  | 22.3  | 23.8  | 26.7  | 29.1  | 30.2  | 31.1  | 30.9  | 30.1  | 27.8  | 24.9  | 20.7 |
| Mean Daily<br>Temperature<br>Range                                            | MDBR        | MDBR   | 7.8   | 7.2   | 6.8   | 6.6   | 7.0   | 6.9   | 7.6   | 7.8   | 7.9   | 8.5   | 8.6   | 8.6  |
|                                                                               |             | MCDBR  | 10.1  | 9.6   | 8.8   | 8.3   | 8.7   | 8.5   | 9.3   | 9.8   | 9.5   | 9.5   | 10.0  | 10.4 |
|                                                                               |             | MCWBR  | 5.0   | 4.5   | 4.1   | 3.3   | 3.3   | 2.8   | 2.8   | 3.0   | 3.0   | 3.3   | 4.0   | 4.9  |
|                                                                               | 5% WB       | MCDBR  | 8.2   | 7.7   | 6.8   | 6.8   | 7.4   | 7.4   | 8.0   | 8.3   | 8.3   | 7.7   | 7.7   | 8.0  |
|                                                                               |             | MCWBR  | 4.9   | 4.6   | 4.2   | 3.4   | 3.4   | 3.5   | 3.0   | 3.0   | 3.1   | 3.0   | 3.0   | 4.0  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.606  | 0.667 | 0.801 | 0.783 | 0.629 | 0.592 | 0.576 | 0.640 | 0.672 | 0.699 | 0.592 | 0.576 |      |
|                                                                               | taud        | 1.720  | 1.646 | 1.446 | 1.510 | 1.831 | 1.931 | 1.968 | 1.792 | 1.708 | 1.609 | 1.807 | 1.803 |      |
|                                                                               | Ebn at Noon | 650    | 645   | 586   | 612   | 711   | 732   | 744   | 698   | 663   | 612   | 658   | 656   |      |
|                                                                               | Edh at Noon | 213    | 244   | 311   | 296   | 214   | 192   | 185   | 220   | 236   | 250   | 194   | 190   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.86   | 2.86  | 2.98  | 3.23  | 4.02  | 4.22  | 5.04  | 4.84  | 4.57  | 4.22  | 3.42  | 3.12  |      |
|                                                                               | RadStd      | 0.48   | 0.47  | 0.43  | 0.58  | 0.54  | 0.58  | 0.46  | 0.37  | 0.32  | 0.43  | 0.44  | 0.49  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | +0.67   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (15 neighbors) | N/A   | N/A     | N/A    | +0.27   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page